CNENGR115 Computers and Control

Block Diagram Analysis

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Why do we use Laplace transforms for control systems?

- The Laplace Transform (LT) is useful for the study of transient responses (or time responses) of Linear Time-Invariant Systems (LTIS). These are dynamic systems described by linear constant coefficient differential equations (or systems of these) in *continuous* time.

- In almost all introductory control system courses, the real systems are modeled as LTIS (this is an approximate model, but very useful nevertheless) and several properties we wish to control, or evaluate, in the systems are related with transient analysis: static gain, rise time, settling time, overshoot of step-response, ramp-response, frequency response, etc…

- It happens that all those properties/characteristics can be evaluated by using the LT to solve the differential equations. So, to answer to your question, the LT is by far the best mathematical tool to use in control systems; that is, it is the one who provides for a simpler mathematical setup to be solved by the practitioner.
- Other transforms, such as the Fourier Transform (FT) could in principle be used, but the FT is not well suited to the solution of transient response problems in LTIS, it is suited only for frequency response studies, while the LT provides easily for both types of responses.

- Note 1: The transform suited for *discrete* control systems studies is the Z transform; it takes the place of the Laplace Transform which applies to continuous systems.
- Note 2: All real control systems are nonlinear. The LT doesn't apply to nonlinear systems. In these problems several techniques, not based in transforms, are used. There are many of them, suited for several common types of non-linearities. In general, nowadays, after studying and designing a real system LTIS approximation with the LT, one performs *simulations* of a more precise nonlinear model of the system, to check if the linear approximation was close enough to the real thing such that the linear design is still valid.

$$F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-j\omega t} dt$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-\sigma t} e^{-j\omega t} dt = \int_{-\infty}^{\infty} f(t)\, e^{-(\sigma + j\omega)t} dt$$

In CNENGR115…

- For the topics of Fourier and Laplace transforms, students are not required to develop a deep understanding, but must instead achieve a specific level of proficiency.
- You only need to understand that we will use the Laplace transform to solve differential equations in control systems, and have a working knowledge of its basic forms.

Control Lectures

Part 1 – Introduction

Part 2 – Mathematical Models

Part 3 – Control Actions

Transfer Functions (Recap)

- **Laplace Transforms** provide a tool for analytically solving differential equations
- **Transfer Functions** are based on Laplace Transforms with zero initial conditions
- In this module, we use an **operator notation** as follows:

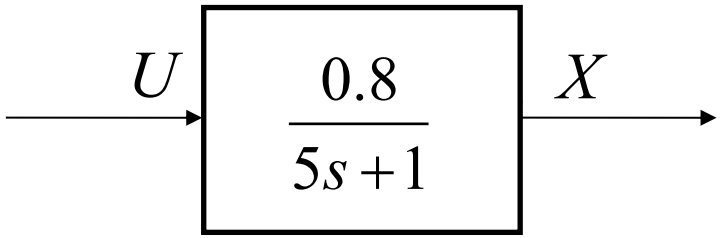
$$\frac{dx}{dt} \rightarrow sX$$

$$\frac{d^2x}{dt^2} \rightarrow s^2X \qquad \frac{d^3x}{dt^3} \rightarrow s^3X$$

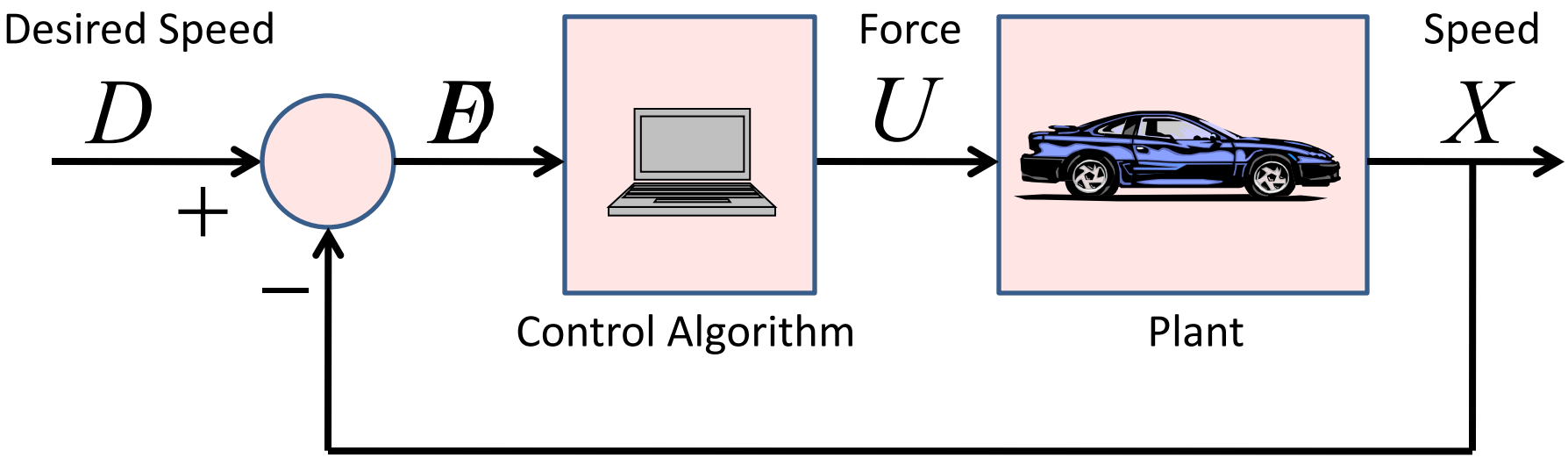
Vehicle Speed Model (Recap)

Assuming $\tau = 5$ and $K = 0.8$ then the model for vehicle speed is:

$$5\frac{dx}{dt} + x = 0.8u \quad \rightarrow \quad 5sX + X = 0.8U \quad \rightarrow \quad (5s+1)X = 0.8U$$

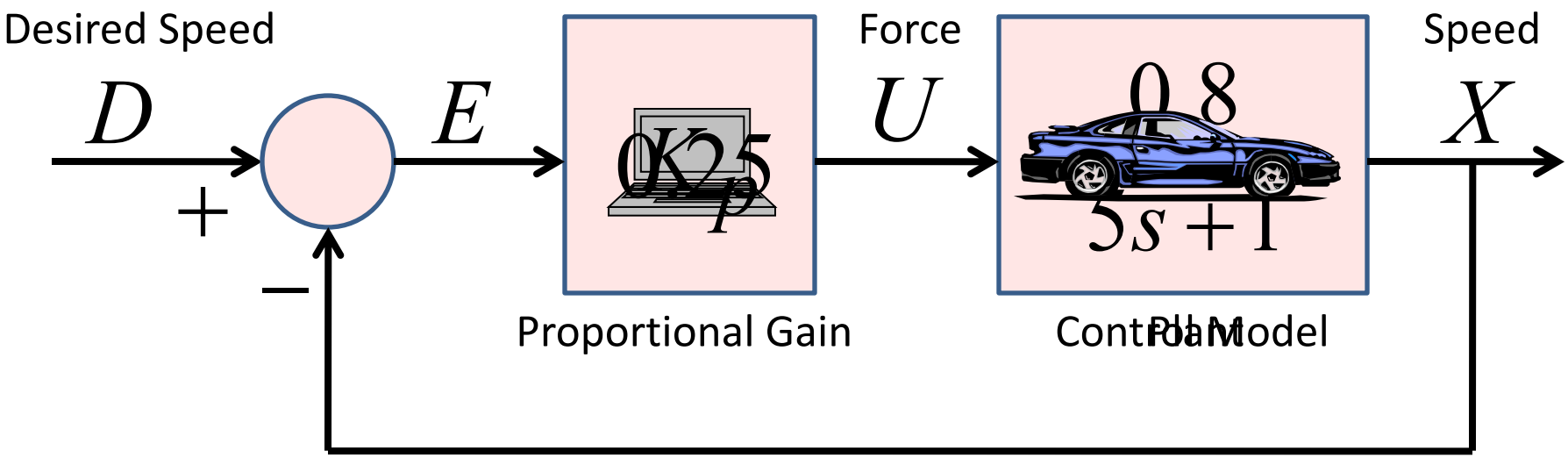


Block Diagram for Closed–Loop Control of Vehicle Speed



$$E = D - X$$

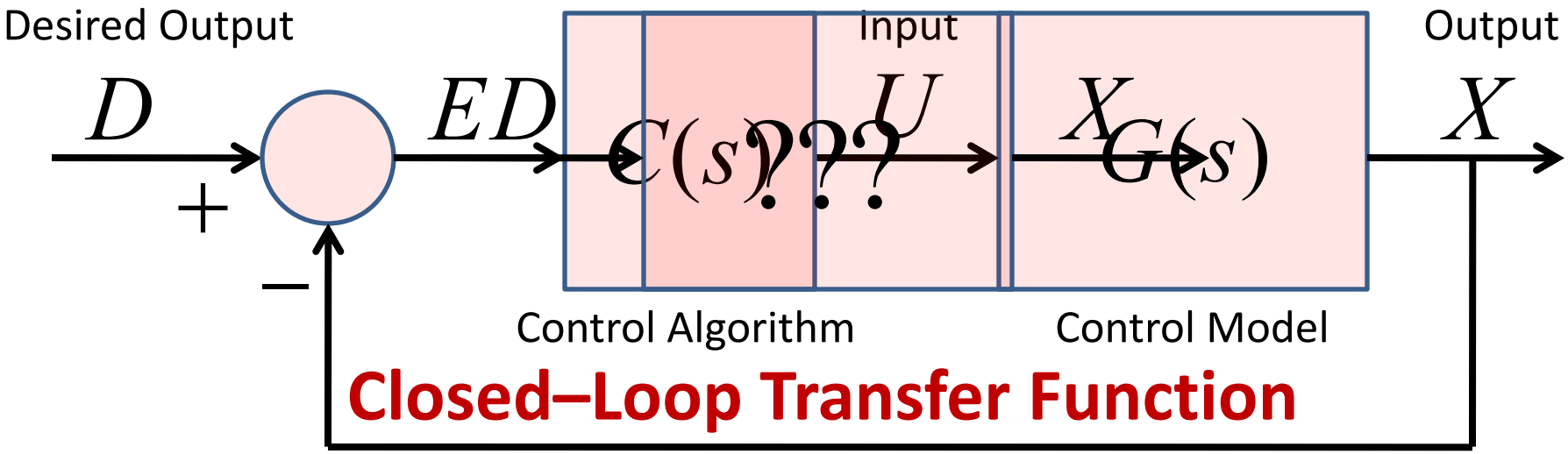
Block Diagram for Closed–Loop Control of Vehicle Speed



Control Algorithm:

$$U = K_p E = K_p (D - X)$$

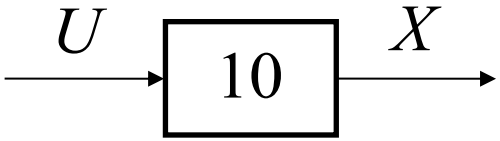
Block Diagram for Closed–Loop Control of a General Transfer Function Model



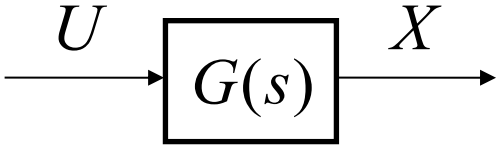
Block Diagram Analysis

- **Block diagrams** provide a straightforward method of analyzing control systems, e.g. to choose appropriate values for the control gain(s)
- We need to learn some rules of block diagram analysis…

Single Transfer Function or Gain

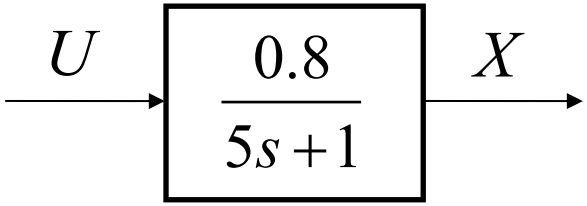


$$X = 10 \cdot U$$



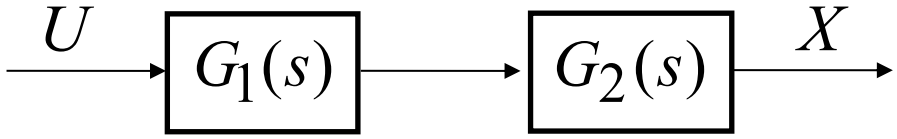
$$X = G(s) \cdot U$$

Single Transfer Function or Gain

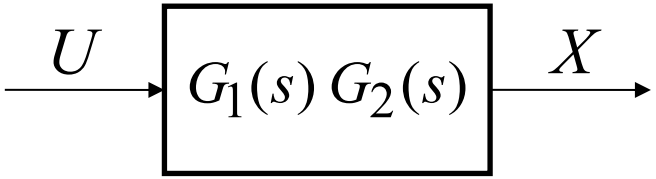


$$X = \frac{0.8}{5s+1}U$$

Two Blocks in Series



$$X = G_1(s)\,G_2(s)\,U = G_2(s)\,G_1(s)\,U$$



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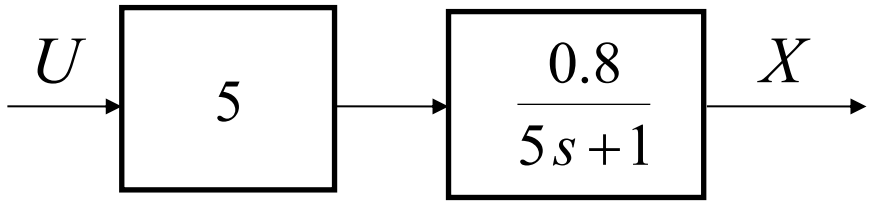

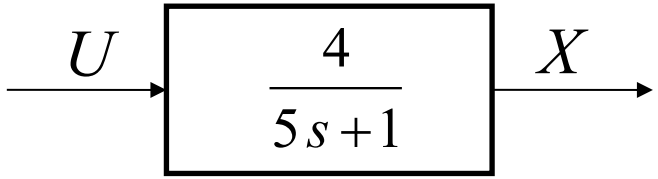

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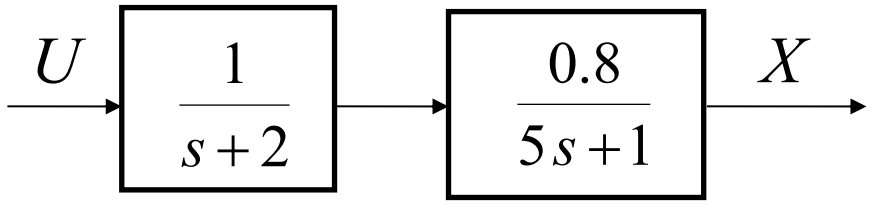
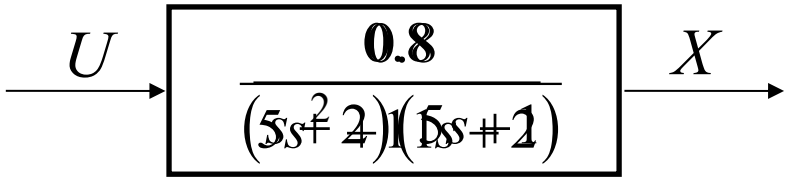
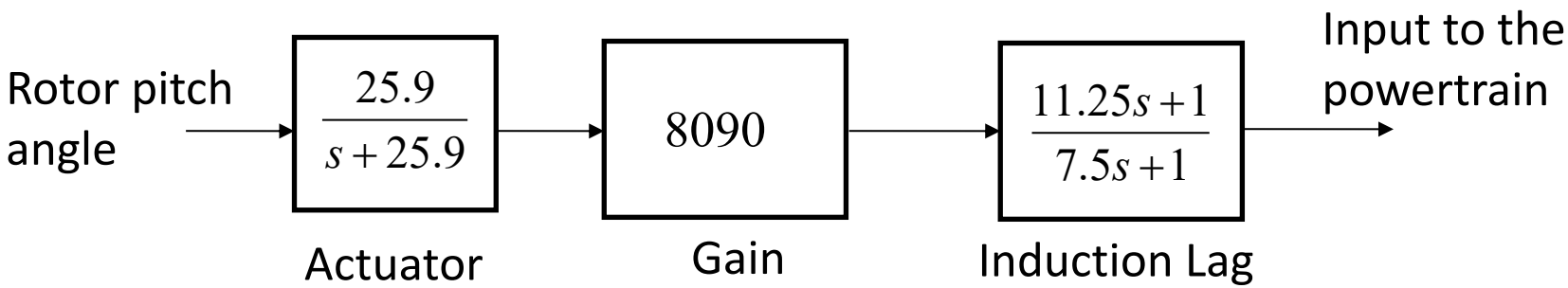
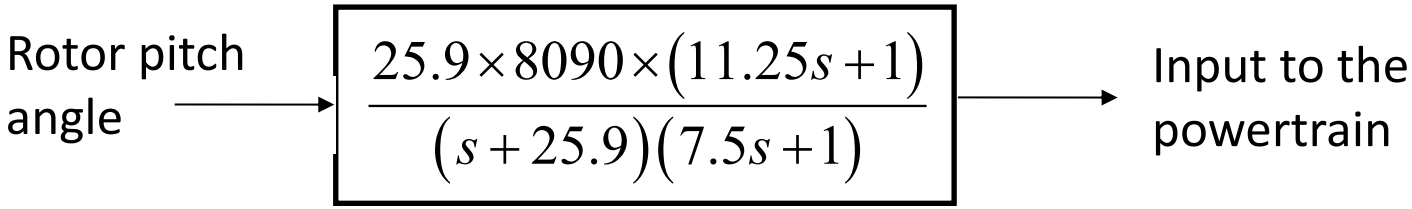




Exercise

- **Wind Turbine Simulation** includes actuator, aerodynamic gain and induction lag in series
- Determine the Transfer Function for the relationship between rotor pitch angle and input to the powertrain





Exercise

Rotor pitch angle

$$\frac{25.9 \times 8090 \times (11.25s + 1)}{(s + 25.9)(7.5s + 1)}$$

Input to the powertrain

$$\frac{2357223.75s + 209531}{7.5s^2 + 195.25s + 25.9}$$

$$\frac{314296.5s + 27937.5}{s^2 + 26.03s + 3.45}$$

Negative Feedback

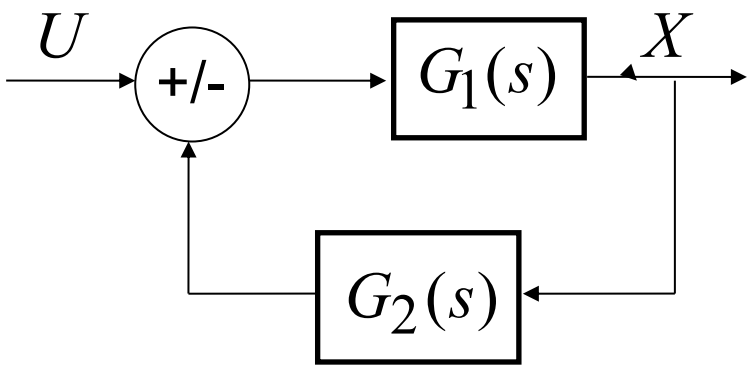
$$X = G_1(s) \times (\text{something})$$

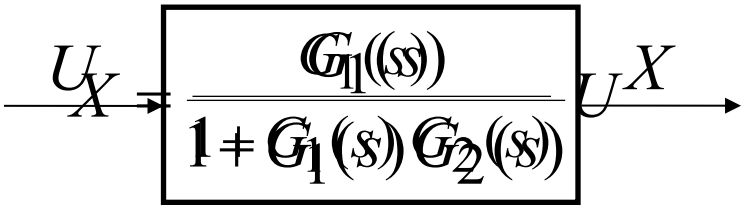
$$X = G_1(s) \times (U - \qquad)$$

$$X = G_1(s)\left(U - G_2(s)X\right)$$

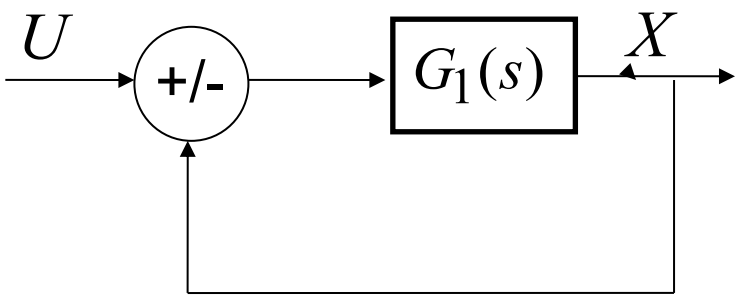
$$X = G_1(s)U - G_1(s)G_2(s)X$$

$$X + G_1(s)G_2(s)X = G_1(s)U$$



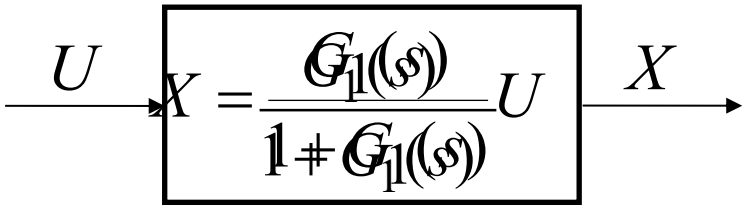


(Unity) Negative Feedback

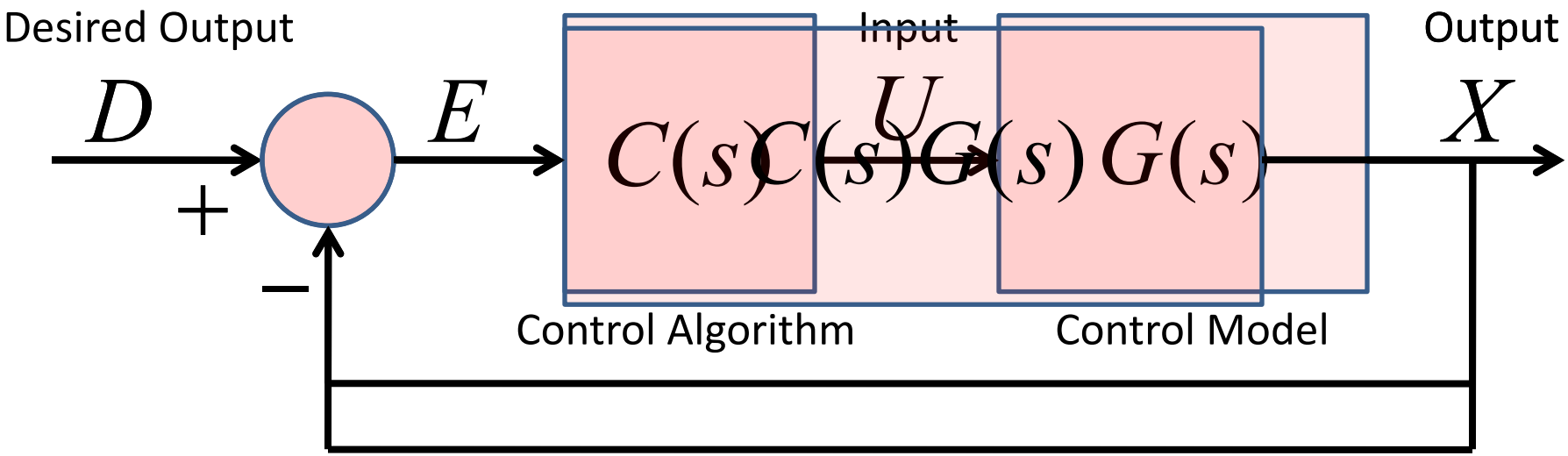


Special case of the previous slide with

$$G_2(s) = 1$$



Block Diagram for Closed–Loop Control of a general Transfer Function Model



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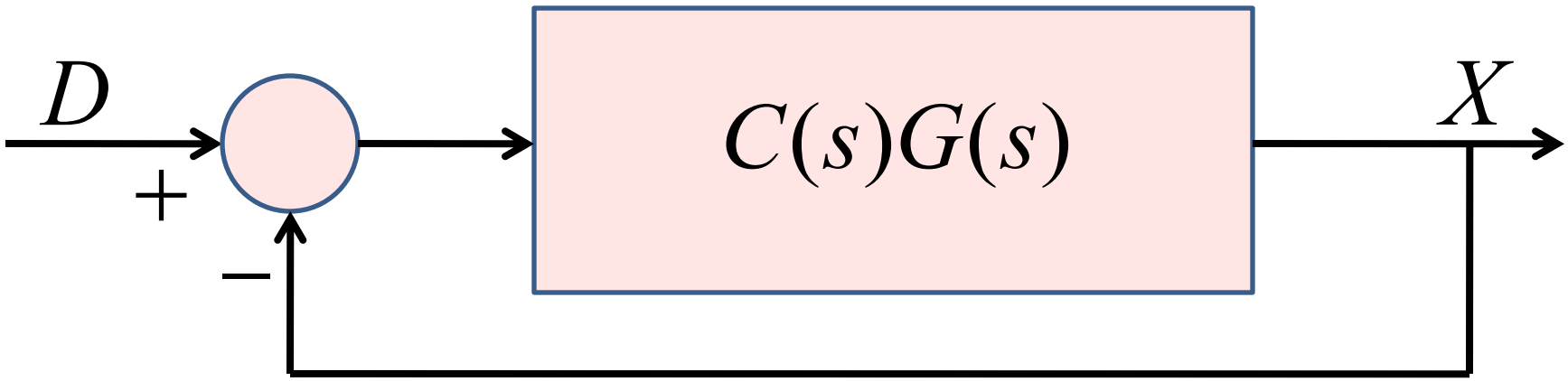

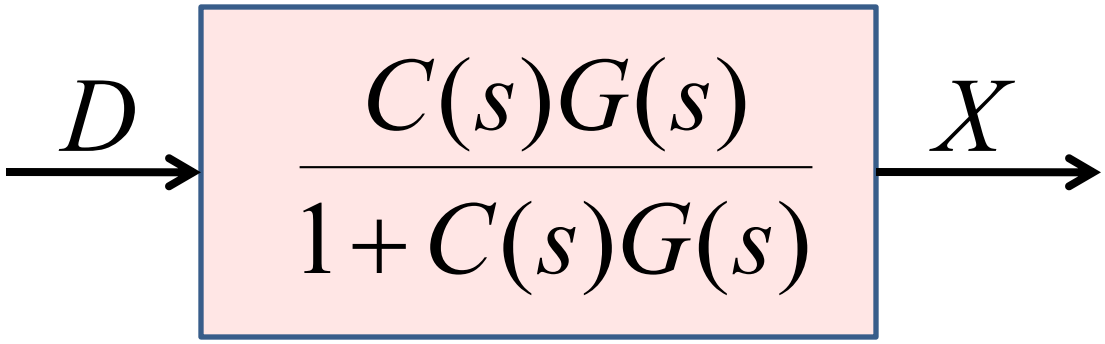

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